Generative models

Michal Pytel Sebastian Trojan

June 16, 2026

Goal and Scope

- Generate new 64×64 cat images from a real cat dataset.
- Compare three model families trained locally:
  - DCGAN,
  - variational autoencoder (VAE),
  - denoising diffusion probabilistic model (DDPM).
- Study selected hyperparameters inside each method.
- Evaluate with FID, sample grids, diversity diagnostics, reconstructions, and interpolation.
- Add an exploratory cats-and-dogs extension using the best cat-only settings.

Dataset and Preprocessing

- Cat dataset: 9997 images.
- Mixed cats-and-dogs dataset: 25000 training images.
- Input images had different sizes and aspect ratios.

Preprocessing

1. Convert to RGB.
2. Centre-crop to square.
3. Resize to 64×64.
4. Random horizontal flip.
5. Normalise to $[-1, 1]$.

The low resolution was chosen to make all experiments feasible on limited compute.

Compared Models

DCGAN

- Latent vector $\mathbf{z}$
- Convolutional generator/discriminator
- Least-squares GAN loss
- Label smoothing and instance noise

VAE

- Encoder and decoder
- Gaussian latent vector
- Reconstruction plus KL loss
- Temperature-controlled sampling

DDPM

- U-Net denoiser
- Learns to predict noise
- Iterative reverse process
- Starts generation from Gaussian noise

Evaluation Protocol

- Fréchet Inception Distance (FID) compares Inception feature statistics:
$$\mathrm{FID} = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right)$$
- DCGAN and VAE FID: 5000 generated images.
- DDPM FID: 2500 generated images because sampling is iterative and slower.
- Qualitative checks: sample grids, VAE reconstructions, and interpolation.
- Diversity diagnostics: pairwise RMSE and pixel standard deviation.

Hyperparameters Tested

Method	Main factors varied
DCGAN	discriminator learning rate, generator learning rate, latent dimension, generator width, instance noise, label smoothing, minibatch-standard-deviation feature
VAE	KL strength, prior-matching variant, latent dimension, decoder refinement, multiscale reconstruction loss, sampling temperature
DDPM	U-Net width, noise schedule, learning rate, dropout, number of diffusion timesteps

- Each model family was compared internally by changing one main factor at a time.
- All variants were trained for 300 epochs.
- FID was interpreted together with generated grids and diversity diagnostics.

Best Hyperparameter Results

DCGAN

Variant	FID
D LR 10^{-4}	**143.79**
G width 96	153.95
No noise	155.60
Baseline	161.62

VAE

Variant	Temp.	FID
$\beta = 0.5$	0.65	**176.68**
Prior	0.50	189.33
$\beta = 0.25$	0.50	191.54
Extra decoder	0.50	193.75

DDPM

Variant	FID
Width 96	**48.00**
Baseline	48.90
Cosine	64.18
LR 10^{-4}	70.61

The best model inside each family was used for the final comparison.

VAE Temperature Results

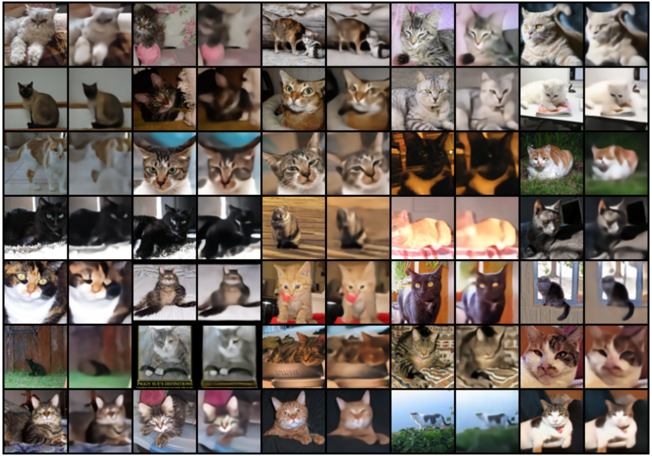
Temp.	Mean FID $\pm$ std.	Best FID
0.50	**209.25** $\pm$ **24.10**	188.23
0.65	209.89 $\pm$ 20.78	**176.68**
0.80	230.99 $\pm$ 14.74	199.93

- 0.50 keeps latent samples closer to the prior centre: smoother but less varied.
- 0.65 gives a moderate amount of variation while keeping more cat-like structure.
- 0.80 explores farther latent regions, which increases malformed textures and anatomy.

Temp. 0.50 Temp. 0.65 Temp. 0.80

Same seed and $\beta = 0.5$ model.

VAE Reconstructions



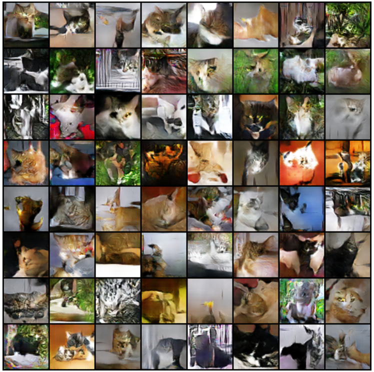
Reconstructions preserve pose and colour better than random samples, but texture and edges are smoothed.

Final Quantitative Comparison

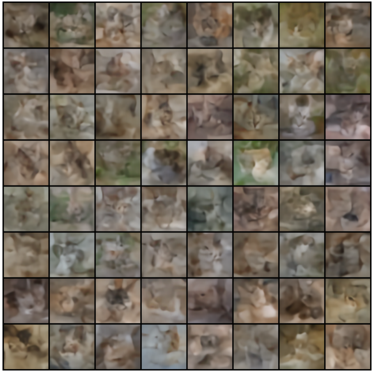
Method	FID	Diversity RMSE	Pixel std.
DCGAN	143.79	0.3682	0.2644
VAE	176.68	0.1351	0.0966
DDPM	**48.00**	**0.4179**	**0.3084**

- DDPM had the strongest FID and diversity diagnostics.
- DCGAN produced sharper images than VAE, but more distorted outputs.
- VAE was stable and reconstructed real cats well, but random samples were blurry.

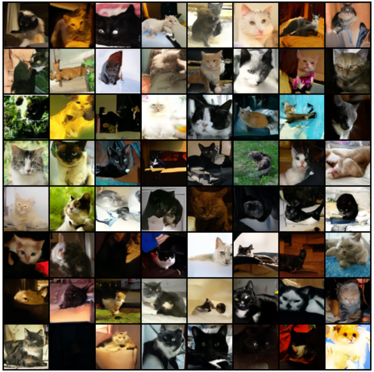
Qualitative Cat-Only Results



DCGAN



VAE



DDPM

DDPM produced the most recognisable and varied cat samples; VAE samples were the smoothest.

Interpolation Comparison

DCGAN

VAE

DDPM

Each row shows 10 points: two endpoints and eight linear interpolations. DDPM gave the smoothest semantic transition; VAE was smooth but blurry, and DCGAN was less stable in the middle samples.

Mode Collapse and Diversity

- Mode collapse was checked with sample grids and pixel-space diversity diagnostics.
- Final diversity RMSE:
  - DCGAN: 0.3682,
  - VAE: 0.1351,
  - DDPM: 0.4179.
- DCGAN did not collapse to one image type, but repeated textures and malformed samples appeared.

DCGAN stabilisation variants

Variant	FID
Stronger discriminator	**143.79**
No instance noise	155.60
Baseline	161.62
No label smoothing	173.37
Lower generator LR	175.31
Minibatch stddev	183.10

Removing instance noise slightly improved FID, while minibatch stddev did not help in this run.

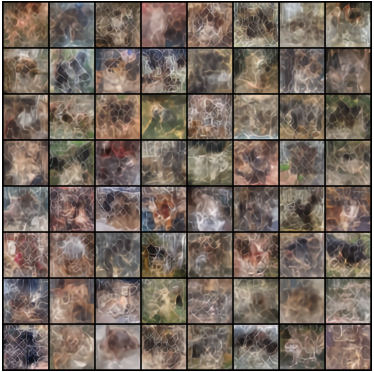
Cats-and-Dogs Extension: FID

Method	FID
DCGAN	131.02
VAE	246.27
DDPM	**74.23**

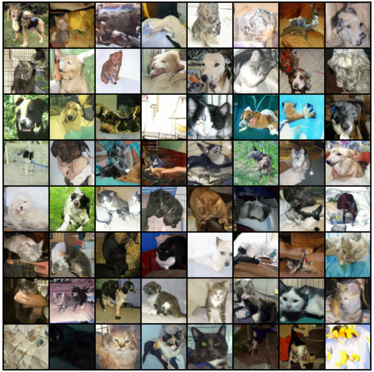
- Same best cat-only settings were reused.
- Models were unconditional: no class label was provided.
- FID was computed against the mixed cats-and-dogs distribution.
- The metric does not prove clean class separation between cats and dogs.

Cats-and-Dogs Extension: Samples

DCGAN



VAE



DDPM

Mixed-data generations include cat-like, dog-like, and ambiguous animal samples.

Conclusions

- The wider DDPM was the strongest model overall.
- DCGAN was a useful lightweight alternative, but less stable.
- VAE reconstructed real images well, but random samples from the prior were weak.
- FID alone was not enough: qualitative grids and diversity checks were necessary.
- The mixed cats-and-dogs experiment confirmed that unconditional generation can create ambiguous samples when classes are combined.

Main takeaway

For this limited-compute setup, DDPM gave the best balance of quality, diversity, and interpolation behaviour.

Thank you